

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Automation and Robotics) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Automation and Robotics) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTMOE505B - RENEWABLE ENERGY SOURCES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 The power from the sun intercepted by the earth is approximately....(A) 1.8×10^8 MW(B) 1.8×10^{11} MW(C) 1.8×10^{14} MW(D) 1.8×10^{17} MW
- 2 A liquid flat plate collector is usually held tilted in a fixed position, facing _____ if located in the northern hemisphere.(A) North(B) South(C) East(D) West
- 3 Maximum efficiency is obtained in....(A) Flat plate collector(B) Evacuated tube collector(C) Line focusing collector(D) Paraboloid dish collector
- 4 Which of the following type of collector is used for low temperature systems?(A) Flat plate collector(B) Line focusing parabolic collector(C) Paraboloid dish collector(D) All of the above
- 5 Which of the following is communally used material for solar cell ?(A) Aluminum (B) Silver(C) Silicon(D) Copper
- 6 In Solar panel, energy conversion is from (A) Solar, Electrical(B) Heat, Light(C) Electrical, Light (D) Electrical, Magnetic
- 7 The amount of energy available in the wind at any instant is proportional to ____ of the wind speed. (A) Square root power of two (B) Square root power of three(C) Square power(D) Cube Power
- 8 Why is wind turbine designed to stop operation at cut out velocity? (A) To protect wheel against damage (B) To make a quick stop in emergencies (C) To improve the efficiency (D) In order to adjust the blades to wind direction
- 9 The aerobic digestion of sewage is used to produce _____ (A) Biomass (B) Biofuels(C) Synthetic fuels (D) Metal articles
- 10 _____ is called as the biogas.(A) Bio ethanol (B) Bio methane(C) Bio diesel (D) Bio butanol
- 11 What does OTEC stand for?(A) Ocean thermal energy cultivation (B) Ocean thermal energy conversion(C) Ocean techno energy conservation(D) Ocean thermal energy consumption
- 12 Nuclear reactors usually employ(A) fission.(B) fusion(C) both fission and fusion. (D) none of the above.

Q2. Solve the following.

- A) Explain in detail energy reserves in India 6
- B) Explain solar Geometry With neat sketch. 6
- Q3. Solve the following.
- A) Explain with neat sketch construction of flat plate collectors & list out its limitation 6
- B) Explain with neat sketch Evacuated Tube Collectors, Advantages Limitations 6
- Q4. Solve Any Two of the following.
- A) Explanation of the components and working principles of heliostat concentrator 6
- B) Define a) Declination angle b) Solar Azimuth angle c) Altitude angle. 6
- C) Explain Solar drying system with advantages & disadvantages. 6
- Q5. Solve Any Two of the following.
- A) Enlist the different application of solar energy and explain with neat sketch solar Air Conditioning 6
- B) What is wind energy? Give classification of types of windmills. 6
- C) Explain selection of site for wind turbine generation system. 6
- Q6. Solve Any Two of the following.
- A) What is OTEC system? Explain working of open cycle OTEC system. 6
- B) What is biomass energy? Explain with neat sketch floating drum type 6
- C) Explain hydroelectric power plant with neat sketch 6

***** End *****